

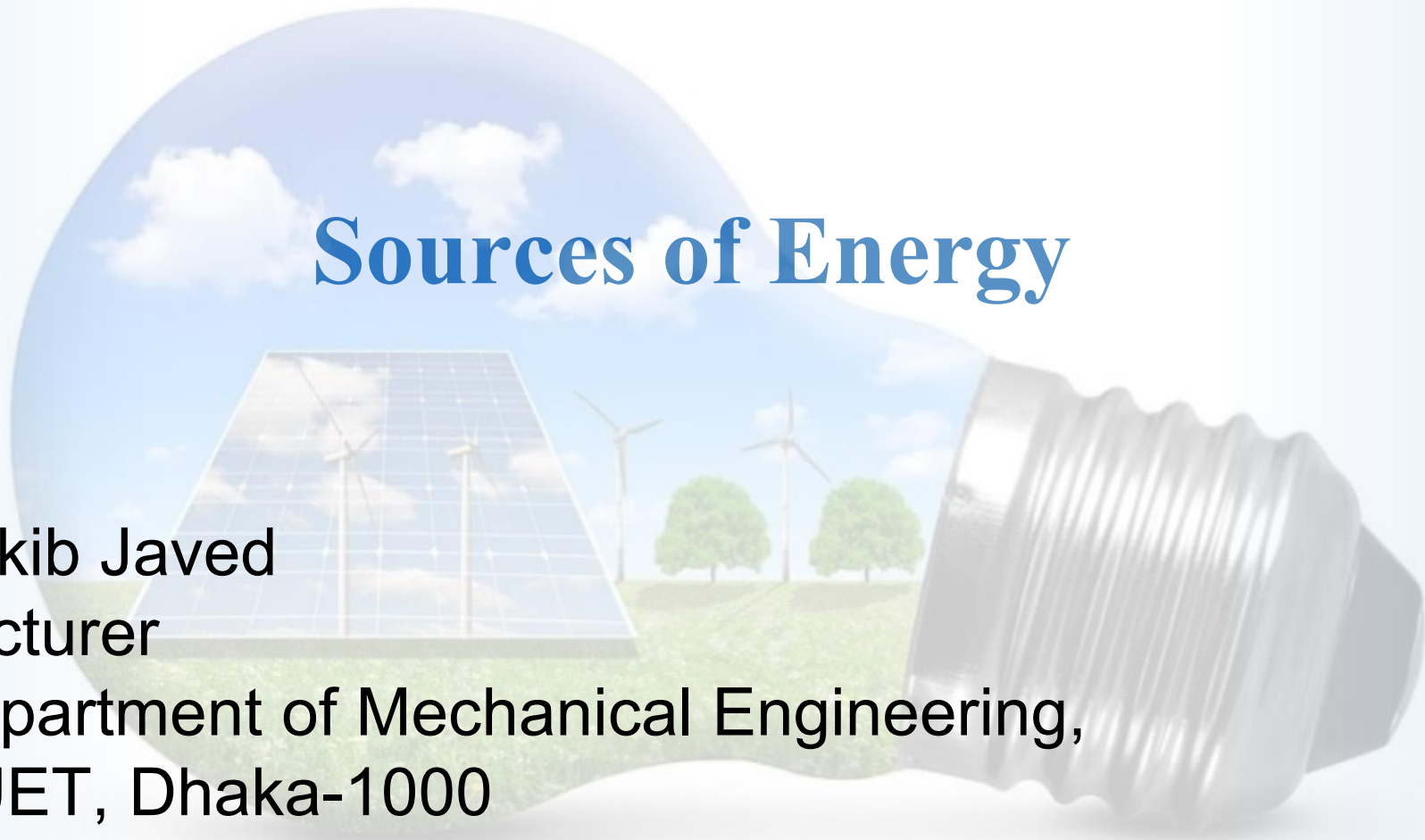
ME 165

Basic Mechanical Engineering

Sources of Energy

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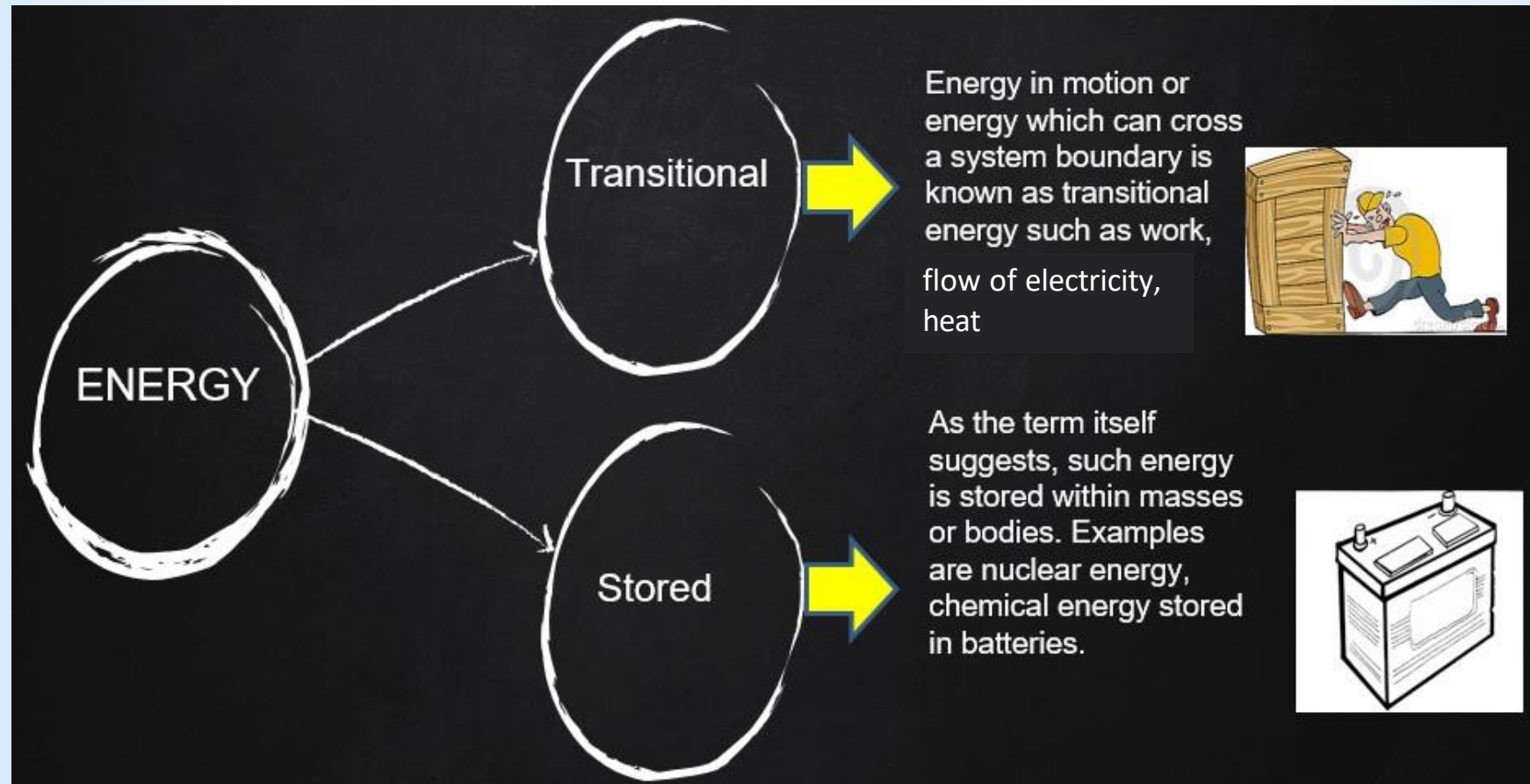


What is Energy

- Energy is the capacity of a physical system to perform work.
- We know from the Law of Conservation of Energy, that energy can neither be created nor destroyed but merely converted from one form to another.
- For example, water falling from a height and rotating a turbine utilizes the initial potential energy of the water to kinetic energy of the turbine. The turbine then rotates a generator, converting its kinetic energy to electrical energy, which then reaches our home and is used in numerous ways as we wish.

Classification of Energy

Based on the virtue of the motion of the energy it can be classified as:



Classification of Energy (Contd.)

Energy is also broadly classified into six categories based on the form of energy.

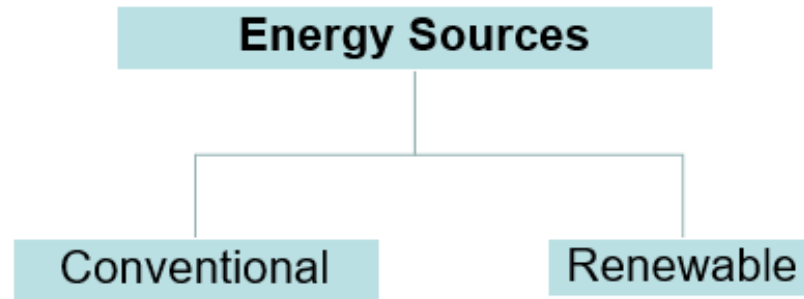
- **Mechanical Energy:** Perhaps the most commonly faced form of energy, it can be defined as the energy acquired by the objects upon which work is done. In other words, the energy that is possessed by an object due to its motion or due to its position. Its transitional form works and can be stored as potential energy (as in water tanks) or as kinetic energy (as in flywheels).
- **Chemical Energy:** It is the energy stored in the bonds of chemical compounds (atoms and molecules) and released during a chemical reaction, usually accompanied by heat. Batteries, petroleum and natural gas are examples of stored chemical energy and a transitional form does not exist.
- **Electrical Energy:** This energy is associated with the accumulation or flow of electrons. The transitional form of electrical energy is electron flow, usually through a conductor of some kind. Electrical energy may be stored as either electrostatic-field energy (electric field produced by the accumulation of charge on the plates of a capacitor) or as inductive-field energy (magnetic field established by the flow of electrons through an induction coil).

Classification of Energy (Contd.)

- **Nuclear Energy:** Just like chemical energy, nuclear energy is a kind of stored energy and is released during reactions occurring within the nucleus of various atoms. Examples include nuclear fission and fusion.
- **Electromagnetic Energy:** Being associated with electromagnetic radiation, this is a pure mass-less form of energy. This radiation exists as only transitional energy traveling at the speed of light.
- **Thermal Energy:** It is the internal energy of an object due to the kinetic energy of its atoms and is dependent on the temperature of the body. Its transitional form is heat while its stored form is internal energy in the form of sensible heat or latent heat.

The 2nd law of thermodynamics states that any other form of energy can be completely converted to thermal energy but thermal energy can never be completely converted to other forms.

Sources of Energy



Conventional sources are fuels which are generally used to produce energy mainly from combustion, chemical reactions or nuclear reactions. Examples include fossil fuels such as coal, petroleum, natural gas etc. and nuclear materials such as U-235.

Renewable sources are an alternative to conventional sources and are naturally replenishing in character. Examples include wind, sunlight, geothermal, tidal, hydroelectric, ocean thermal, biomass and biodiesel etc.

Fossil Fuels

- They were formed many hundreds of millions of years ago from the fossilized remains of dead plants and animals by exposure to heat and pressure in the Earth's crust over this time period.
- Coal, Crude Oil and Natural Gas are the major traditional fossil fuels in use today.



Good to know: Coal is often extracted through strip mining at the surface or deep tunneling. Power shovels, large trucks, bucket wheel excavators, and conveyors are used for this purpose.



Good to know: Oil and natural gas are found much deeper below the surface. Imaging technology is used to examine geological formations for likely oil-bearing areas. At sea, massive drills from ocean platforms are used to bore into the deposits. On land, wells are drilled, and pump jacks extract the oil. Natural gas reserves often exist within oil reservoirs

Fossil Fuels: Advantages

- A major advantage of fossil fuels is their capacity to generate huge amount of electricity in just a single location.
- Fossil fuels are very easy to find.
- When coal is used in power plants, they are very cost effective. Coal is also in abundant supply.
- Transporting oil and gas to the power stations can be made through the use of pipes making it an easy task.
- Power plants that utilize gas are very efficient.
- Power stations that make use of fossil fuel can be constructed in almost any location. This is possible as long as large quantities of fuel can be easily brought to the power plants.

Fossil Fuels: Disadvantages

- Pollution is a major disadvantage of fossil fuels. This is because they give off carbon dioxide when burned thereby causing a greenhouse effect. This is also the main contributory factor to the global warming experienced by the earth today.
- Coal also produces sulfur dioxide, a kind of gas that creates acid rain.
- Environmentally, the mining of coal results in the destruction of wide areas of land. Mining this fossil fuel is also difficult and may endanger the lives of miners.

Coal mining is considered as one of the most dangerous jobs in the world.

What are Fuels

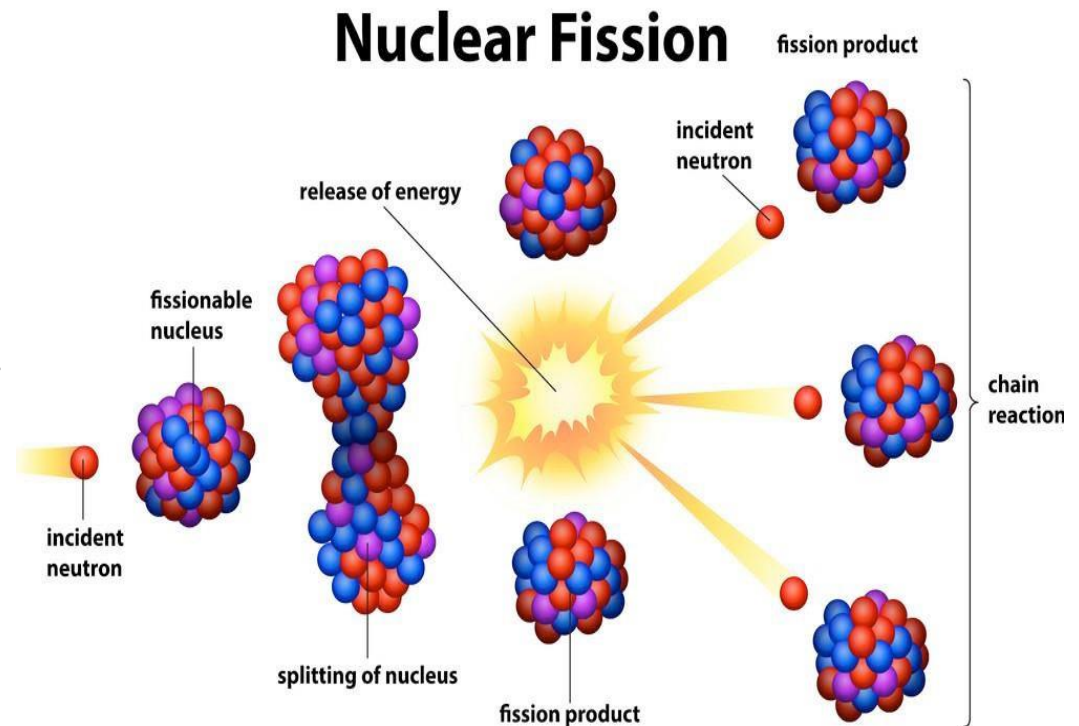
- A Fuel may be defined as a substance (containing mostly carbon and hydrogen) which, on burning with oxygen in the atmospheric air, produces a large amount of heat.
- The **calorific value** (or heating value) of a solid or liquid fuel may be defined as the amount of heat given out by the complete combustion of 1kg of fuel. It is expressed in kJ/kg of fuel or for gases, in kJ/m³ at a specified temperature and pressure.
- Heating value is commonly determined by using a bomb calorimeter.
- There are two types of heating value:
 - Higher Heating Value (HHV): It is determined by bringing all the products of combustion back to the original pre-combustion temperature, and in particular condensing any vapour produced. (Also called gross calorific value or gross energy or upper heating value).
 - Lower Heating Value (LHV): It is determined by subtracting the heat of vaporization of the water vapour from the higher heating value. This treats any H₂O formed as a vapour. The energy required to vaporize the water therefore is not realized as heat. (Also known as net calorific value).

$$HHV = LHV + m_{cw} \times h_v$$

Nuclear Power

Nuclear energy is a kind of stored energy released during reactions occurring within the nucleus of various atoms. Nuclear power plants use low-enriched uranium fuel to produce electricity through a process called fission—the splitting of a fissile atom in a nuclear reactor.

- **Fissionable and Fissile Nuclei:** In nuclear engineering, fissile material is material capable of sustaining a nuclear fission chain reaction. By definition, fissile material can sustain a chain reaction with neutrons of any energy. 'Fissile' is distinct from 'fissionable'. A nuclide capable of undergoing fission (even with a low probability) after capturing a high energy neutron is referred to as 'fissionable'. A fissionable nuclide that can be induced to fission with low-energy thermal neutrons with a high probability is referred to as 'fissile'.



Nuclear Power: Advantages

- The generation of electricity through nuclear energy reduces the amount of energy generated from fossil fuels (coal and oil) therefore lowering greenhouse gas emissions (CO₂ and others).
- Another advantage is the required amount of fuel: less fuel offers more energy. A single uranium pellet, typically 1 cm in diameter and 1 cm in length contains the same energy as a ton of coal.
- The production of electric energy is continuous. A nuclear power plant is generating electricity for almost 90% of annual time
- By reducing the consumption of fossil fuels we also improve the quality of the air affecting quality of human life.

Nuclear Power: Disadvantages

- Despite the high level of sophistication of the safety systems of nuclear power plants the human aspect has always an impact.
- One of the main disadvantages is the difficulty in the management of nuclear waste. It takes many years to eliminate its radioactivity and risks.
- Nuclear plants have a limited life. The investment for the construction of a nuclear plant is very high and must be recovered as soon as possible, so it raises the cost of electricity generated.
- Nuclear power plants generate external dependence. Not many countries have uranium mines and not all the countries have nuclear technology, so they have to hire both things overseas.
- Probably the most alarming disadvantage is the use of the nuclear power in the military industry.

Renewable Energy



Tidal Energy



Wind Energy



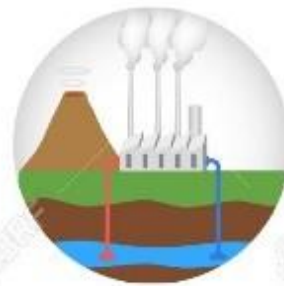
Solar Energy



Renewable energy



Biomass Energy



Geothermal Energy



Hydroelectricity

Solar energy

- The most widely used renewable form of energy, solar energy is usually used both directly or indirectly.
- Indirectly, solar energy affects the wind and ocean thermal energy so affecting the energy obtained from them. In fact, solar energy is one of the most important conditions needed for the formation of fossil fuels.
- Apart from nuclear energy, geothermal energy and tidal power, the sun is responsible for all the forms of energy which are exploited by man.
- Originates with the thermonuclear fusion reactions occurring in the sun. The Solar fusion reaction results in temperature of about 6000°C at the sun's surface. Solar energy represents the entire electromagnetic radiation (visible light, infrared, ultraviolet, x-rays, and radio waves).
- Directly solar energy can be utilized in two ways:
 - Solar thermal energy
 - Solar Photovoltaic

Energy from the sun

- The incident of solar energy flux on the earth's surface is large. The energy flux that strikes the top of the earth's atmosphere is referred to as the solar constant and has a value of 1352 W/m^2 .
- Depending on the time of day and month of year as well as local weather and latitude of a particular location, the amount of insolation that actually reaches the surface will vary from essentially 0 to about 1050 W/m^2 .
- On average, about half of the energy incident on the earth's upper atmosphere makes it to the surface.
- About 21% of the solar flux reaches as direct radiation and about 29% as scattered or diffuse radiation.
- Even with these losses, the amount of solar energy reaches us is significant.
- Energy reaching the earth is incredible. By one calculation, 30 days of sunshine striking the Earth have the energy equivalent of the total of all the planet's fossil fuels, both used and unused!

Solar Thermal Energy

- Application 1: To generate power
 - Step 1: Concentrating the solar energy at a single point or line



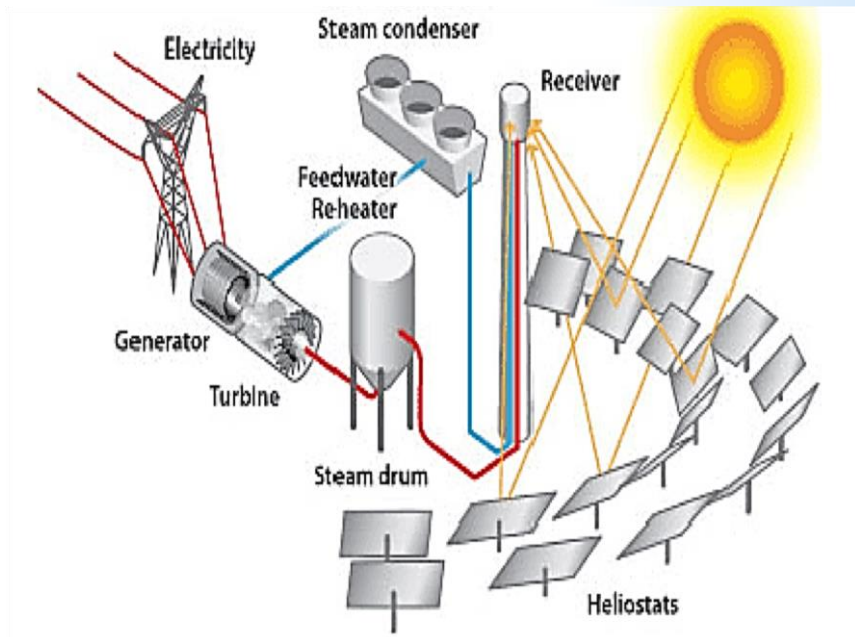
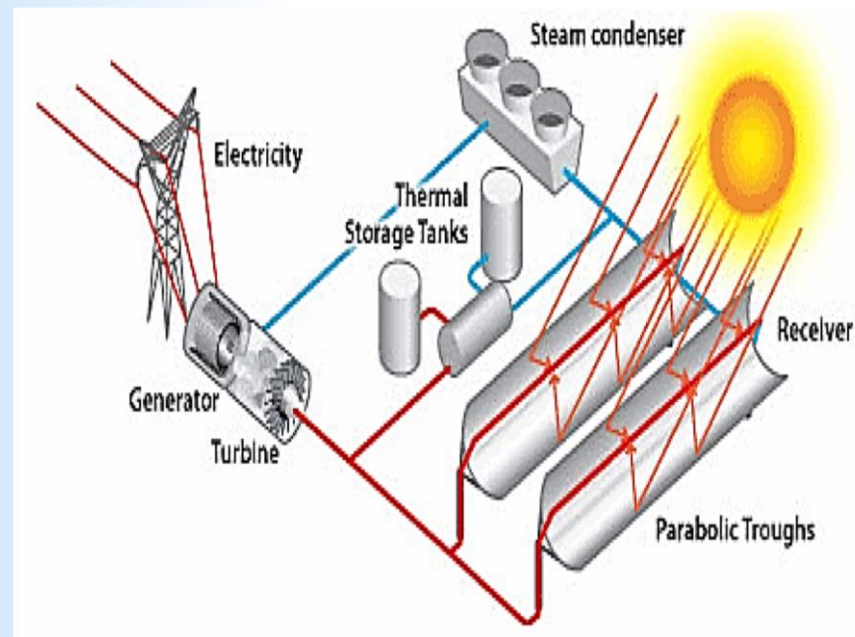
a) Using Parabolic trough collectors



b) Using heliostats for a solar power tower

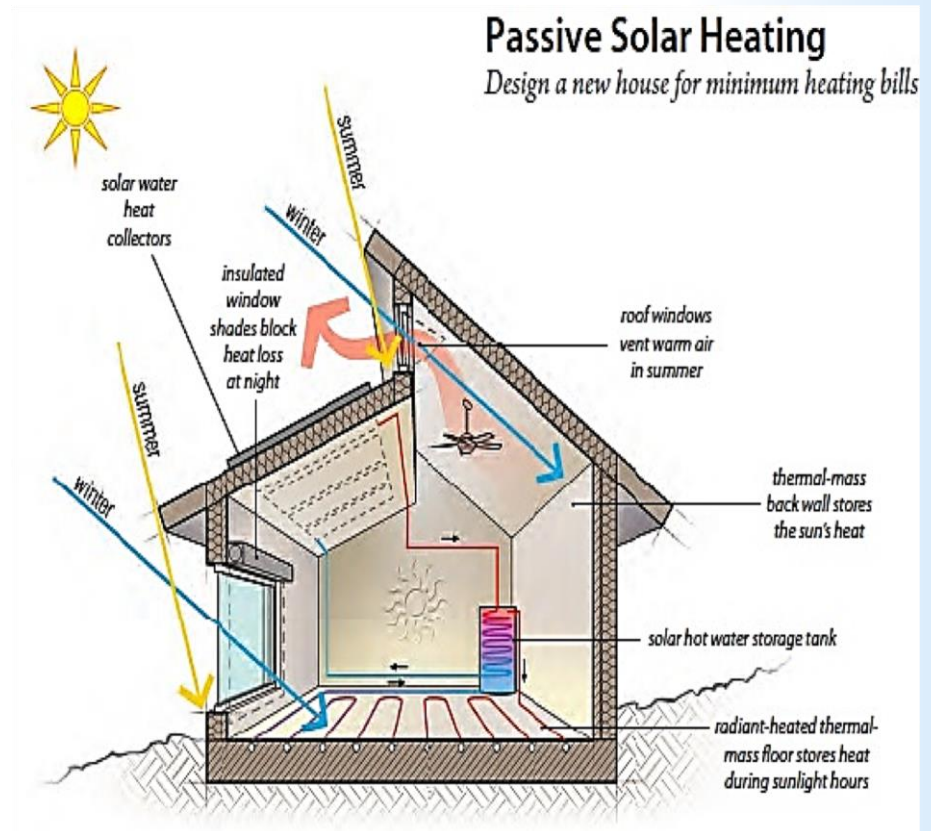
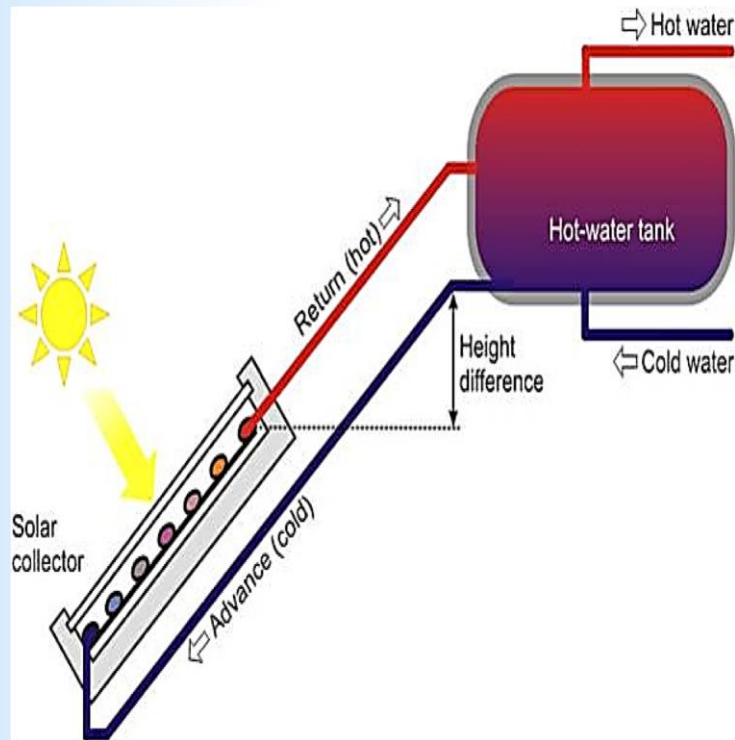
Solar Thermal Energy (Contd.)

- Step 2: Using the heat to generate steam to power a turbine

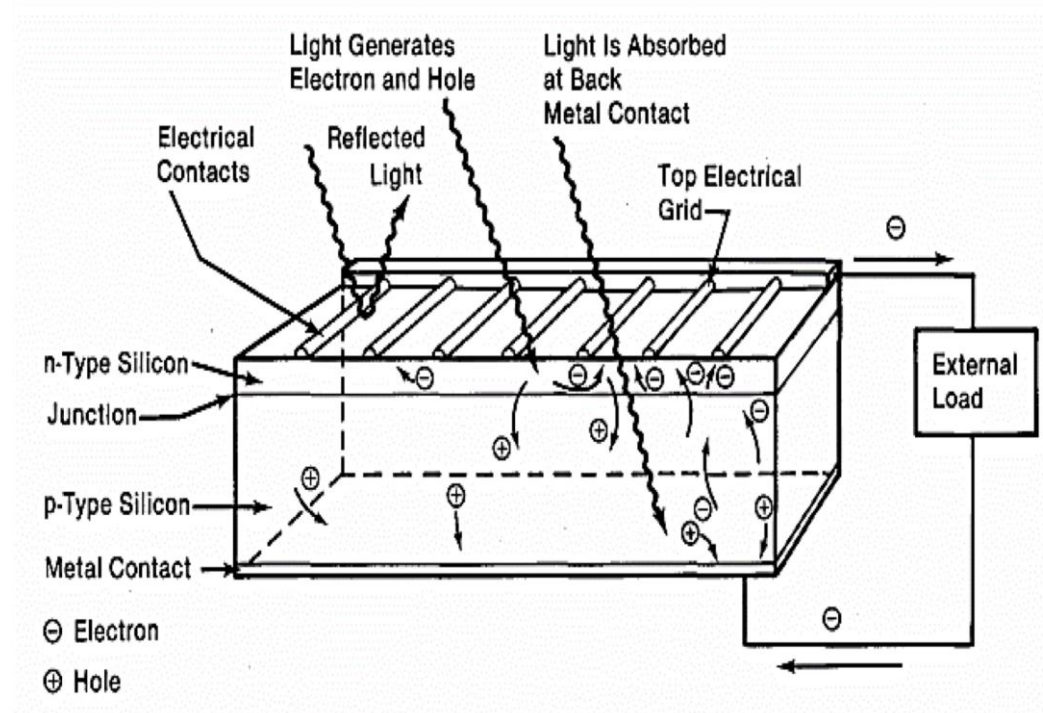


Solar Thermal Energy (Contd.)

➤ Application 2: Solar water heater and/or heating the house

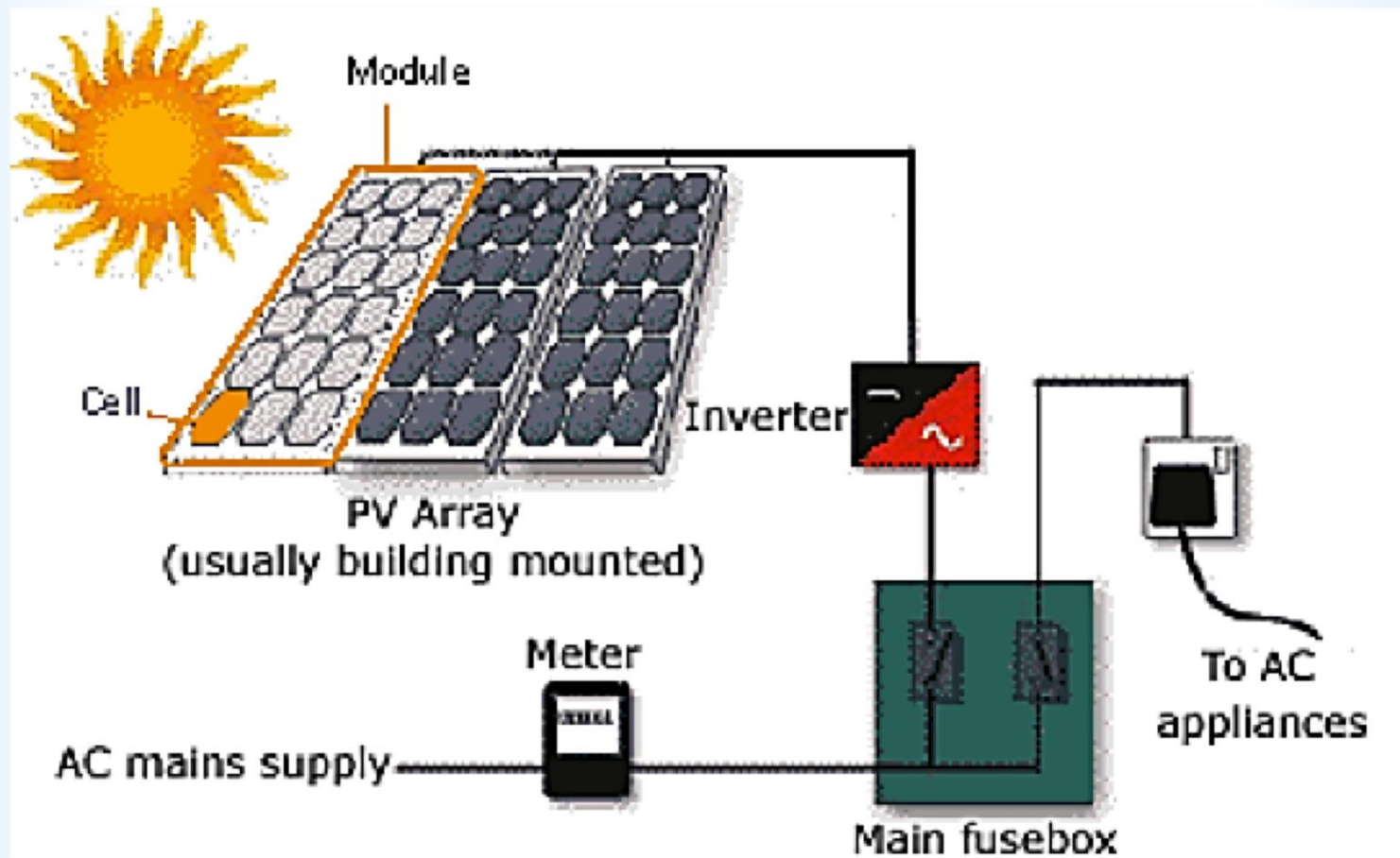


Solar Photovoltaics



Solar cells are composed of various semiconducting materials (mainly silicon) which become electrically conductive when supplied with light or heat, but which operate as insulators at low temperatures. When sunlight strikes a solar cell, an electron is freed by the photoelectric effect. A difference in the electric potentials created, cause electrons to flow and thus current flows when this solar cell is connected to an external load.

Solar Photovoltaics (Contd.)



Solar Energy: Advantages

- Solar power is pollution free and causes no greenhouse gases to be emitted after installation.
- Reduced dependence on foreign oil and fossil fuels.
- Almost no maintenance as solar panels last over 30 years.
- Can be installed almost anywhere; in a field to on a building.
- Can be stored in batteries for use when solar power is not available.
- Energy source is significantly large.

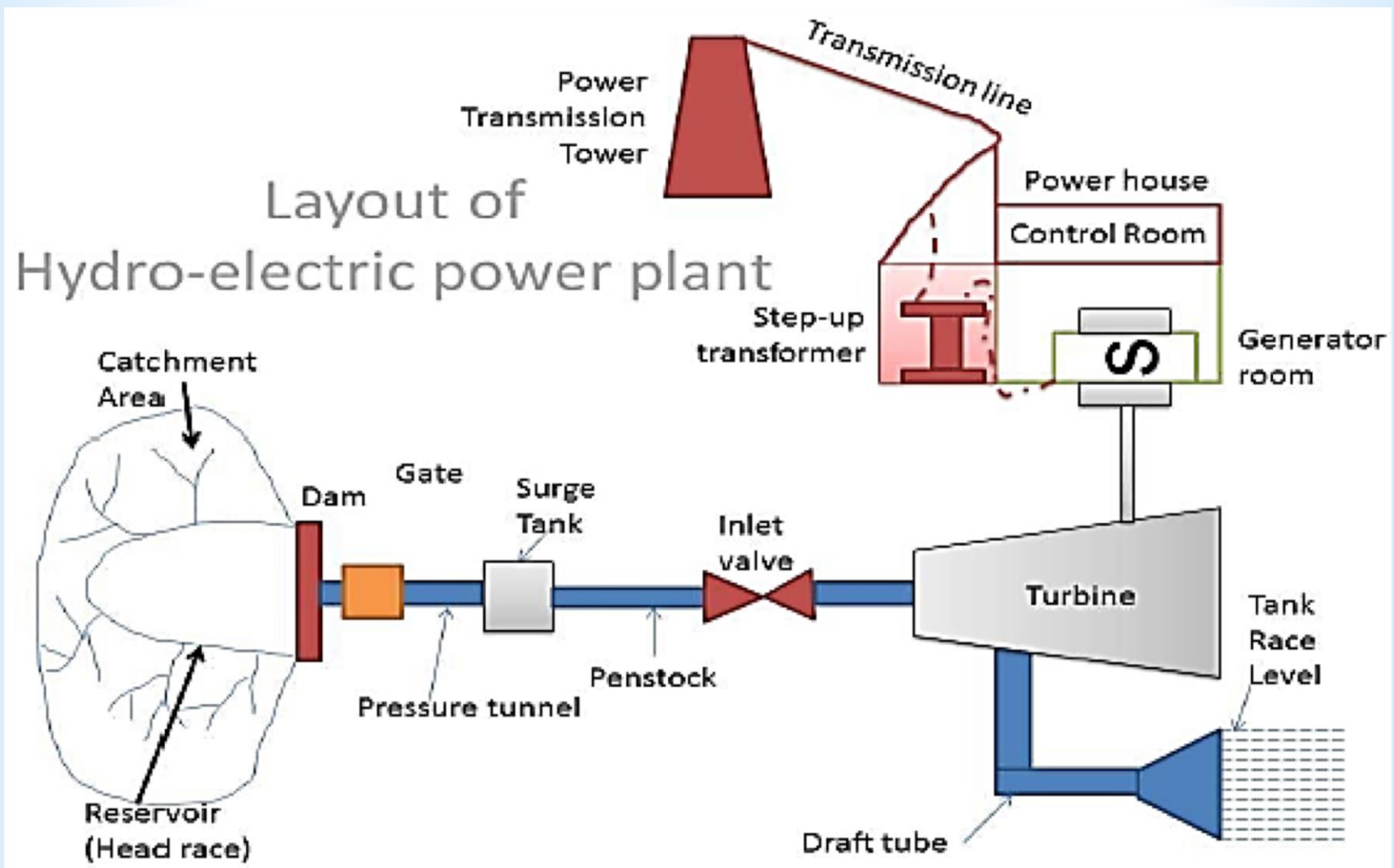
Solar Energy: Disadvantages

- High initial costs for material and installation
- Needs lots of space
- No solar power at night so there is a need for a large battery bank
- Depending on geographical location the size of the solar panels vary for the same power generation
- Cloudy days do not produce much energy

Hydroelectric Energy

- Hydroelectricity is the term referring to electricity generated by hydropower; the production of electrical power through the use of the gravitational force of falling or flowing water.
- The theory is to build a dam on a large river to store a lot of water behind it in the reservoir. Near the bottom of the dam wall, there is the water intake. Water flows from the reservoir through a pipeline to the turbine propeller and makes it rotate. The shaft of the turbine is coupled with the shaft of the generator, resulting in the production of electric power. Power lines are connected to the generator that carries electricity.

Layout of hydroelectric power plant



Hydroelectric Energy (Good to know)

- About 20% of total world electric power is generated using hydropower
- In two countries (Norway & Brazil), more than 80% of the nation's power requirement is met by hydropower
- As per World Power Organization (WPO), world hydro potential is 5000 GW whereas about 4-6% (200-300 GW) is currently developed
- No Green House Gas (GHG) emission
- The 22,500 MW Three Gorges hydroelectric power plant in Yichang, Hubei province, China, is the largest hydropower station in the world exploiting the water resource of the Yangtze River (1993-2012).

(<https://www.youtube.com/watch?v=-87IN7CVTkU>)

- The Kaptai Hydroelectric power plant is the only hydropower plant in Bangladesh located on the Karnaphuli River at Kaptai, Chittagong. The installed capacity is 230MW.

Wind Energy

- Wind energy can be extracted by wind turbines, windmills, windpumps.
- A wind turbine (WT) converts kinetic energy from the wind into electrical power.
- A windmill is a mill that converts the kinetic energy of wind into rotational energy for grinding grains.
- A windpump uses the kinetic energy of wind for pumping water.



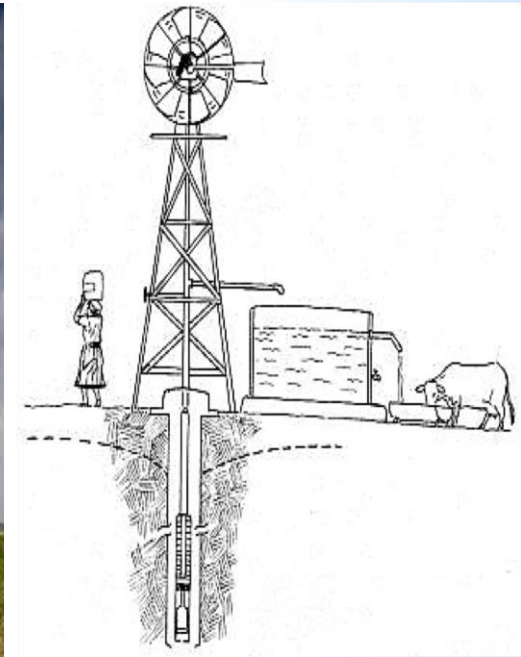
Horizontal axis WT



Vertical axis WT



Windmill

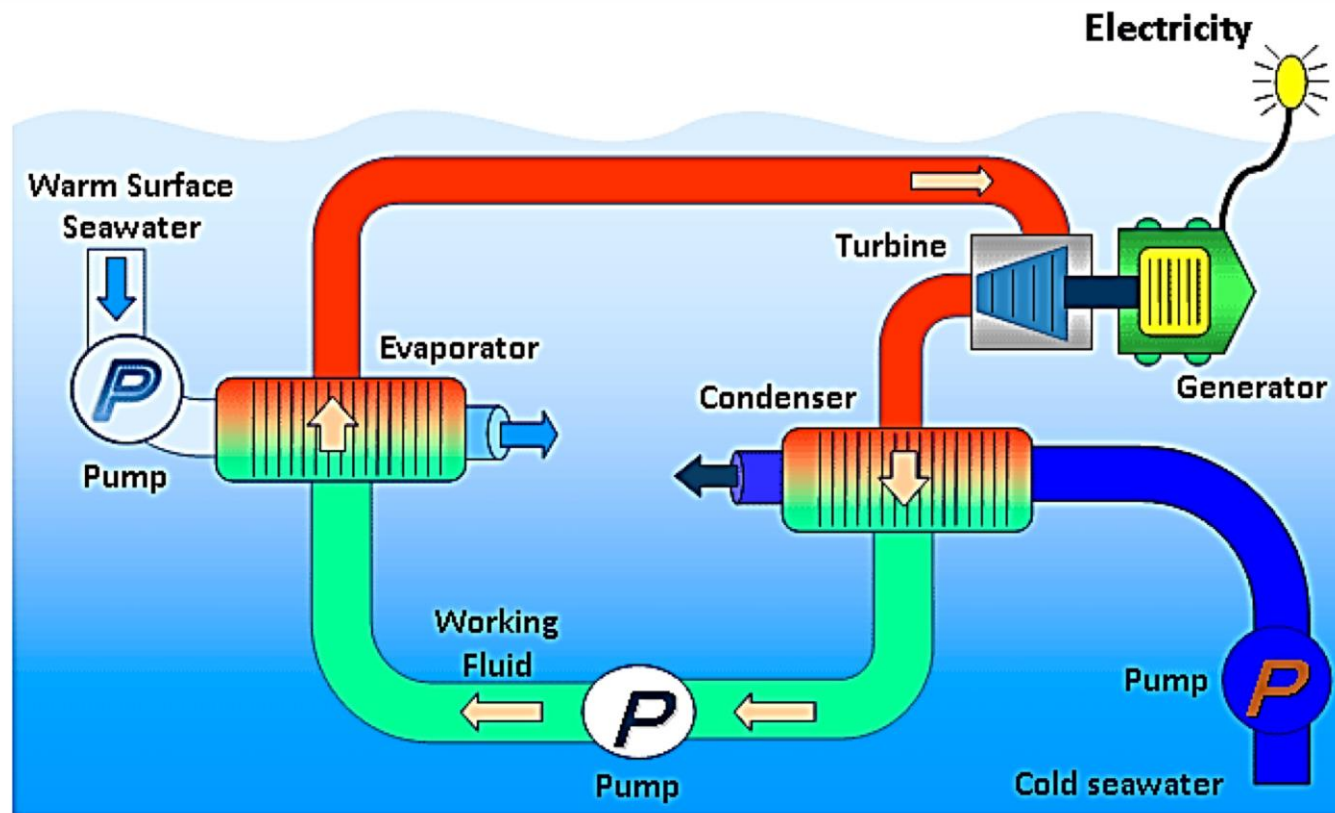


Wind pump

Ocean Thermal Energy

- Ocean Thermal Energy Conversion (OTEC) is a marine renewable energy technology that uses the solar energy absorbed by the oceans.
- The basic working principle of OTEC is quite simple. The warm water is used to evaporate a working fluid with a low boiling point. The high pressure vapor that is produced drives a turbine-generator to produce electricity. The cold deep seawater is used to condense the working fluid vapor back into a liquid.
- One of the main advantages of OTEC with other renewables, such as wind and solar energy, is the fact that OTEC is an energy source available day and night.

Ocean Thermal Energy (Contd.)



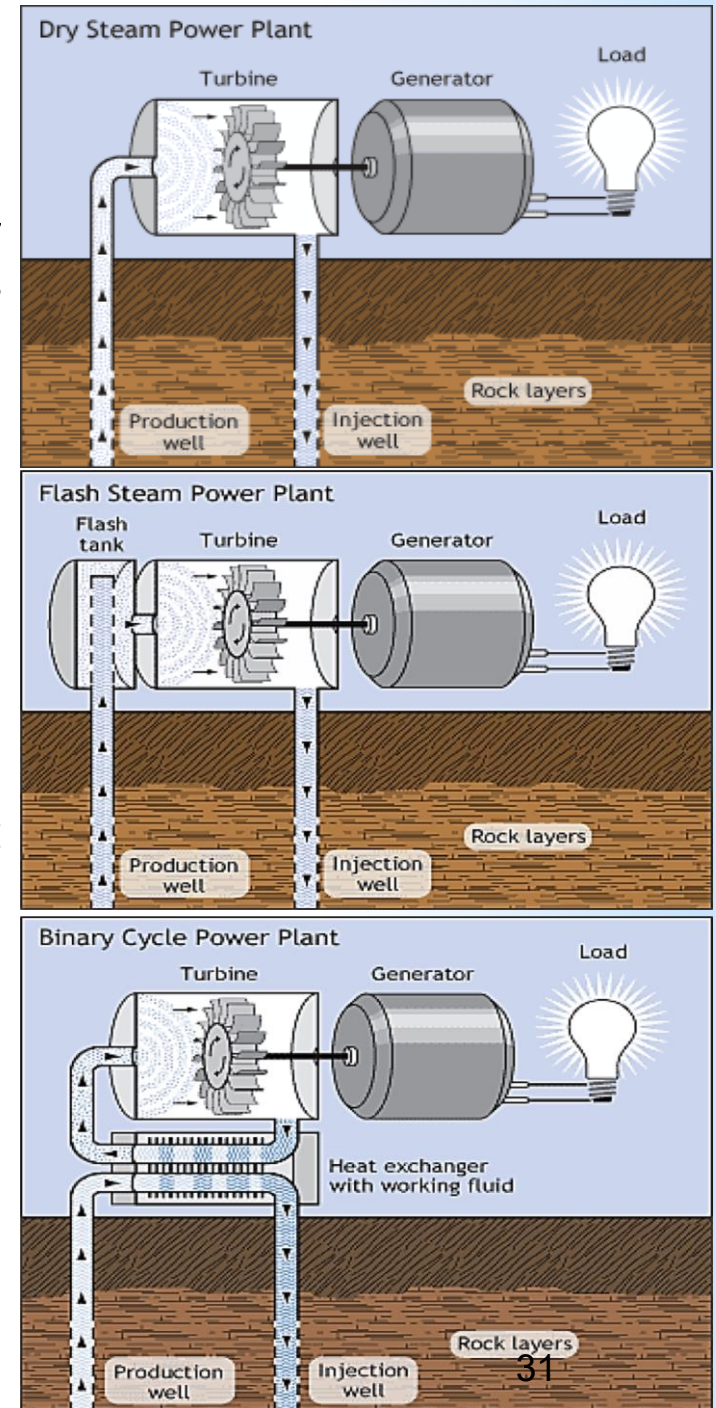
1. Warm seawater and Cold seawater are pumped to the evaporator and condenser.
2. The cold working fluid is pumped to the evaporator
3. The evaporator uses the warm sea water to vaporise the working fluid
4. The vapor rotates the turbine thereby generating electricity
5. The vapor then enters the condenser where the cold seawater cools it to a fluid
6. The fluid returns to the pump and the cycle is repeated

Geothermal Energy

- The center of the Earth is around 6000 degrees Celsius -hot enough to melt rock. Even a few kilometers down, the temperature can be over 250°C. In general, the temperature rises one degree Celsius for every 36 meters you go down. In volcanic areas, molten rock can be very close to the surface and in such areas geothermal energy has been used for thousands of years for cooking and heating.
- There is more than 11,000 megawatts of geothermal energy already operating around the world. It is also another renewable energy which can generate electricity day and night.
- There are three designs for geothermal power plants, all of which pull hot water and steam from the ground, use it, and then return it as warm water to prolong the life of the heat source.

Geothermal Energy (Contd.)

- In the simplest design, the steam goes directly through the turbine, then into a condenser where the steam is condensed into water. This system is used when steam is directly available.
- In a second approach, very hot water is depressurized or "flashed" into steam which can then be used to drive the turbine. This system is used when very hot water is available.
- In the third approach, called a binary system, the hot water is passed through a heat exchanger, where it heats a second liquid—such as isobutane—in a closed loop. The isobutane boils at a lower temperature than water, so it is more easily converted into steam to run the turbine. This system is currently being focused on as more hot water sources are available.

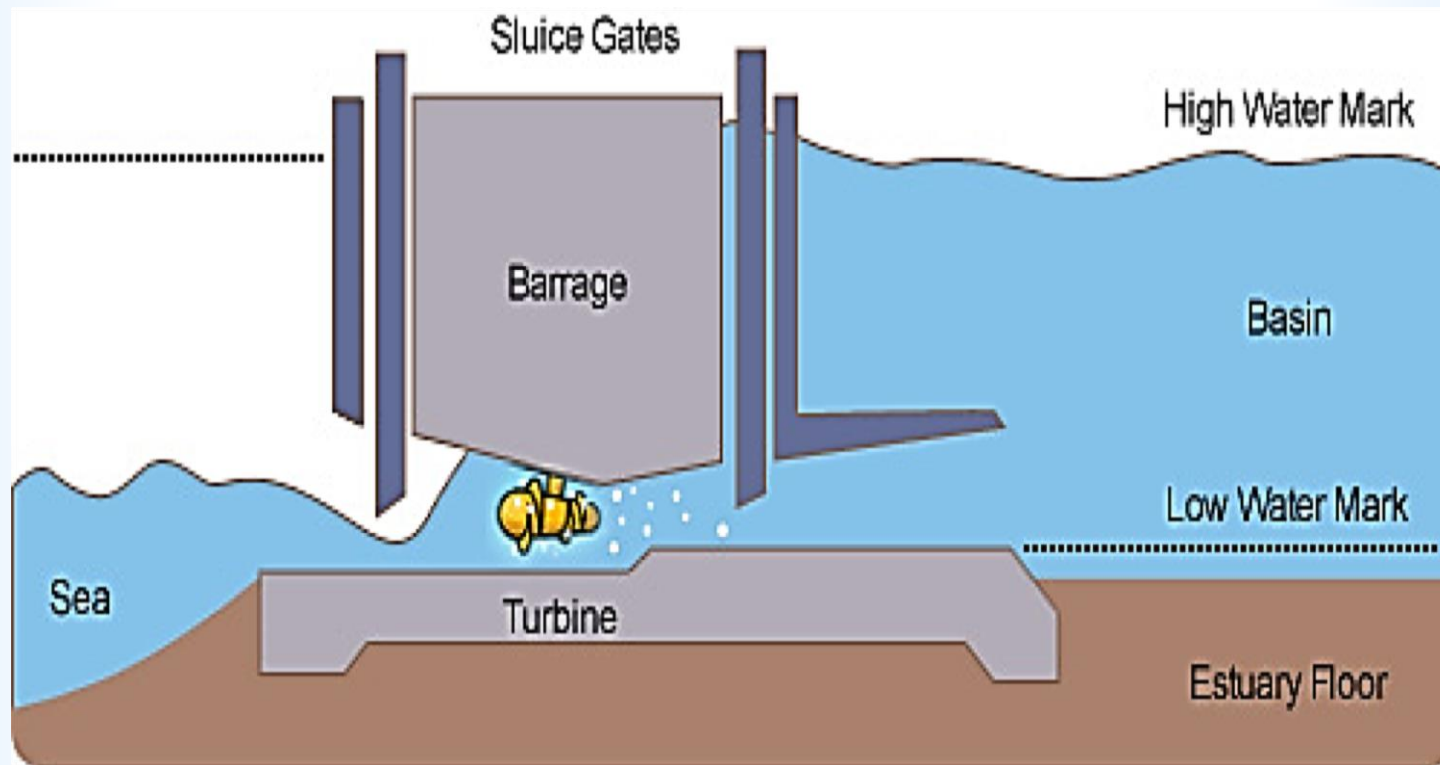


Tidal Energy

Tidal energy or tidal power is a form of renewable energy obtained due to alternating sea levels. The kinetic energy from the natural rise and fall of tides is harnessed and converted into electricity. Tides are caused by the combined gravitational forces of the moon, sun, and earth. However, tides are influenced most by the moon. The moon's gravitational force is so strong that it tugs the ocean into bulge. The high and low tides create tidal currents, which are essential in generation of this kind of energy mostly prevalent in coastal areas. Tidal energy is considered a renewable energy resource because the oceans and seas will remain until the end of time and tides are highly predictable.

Tidal Energy (Contd.)

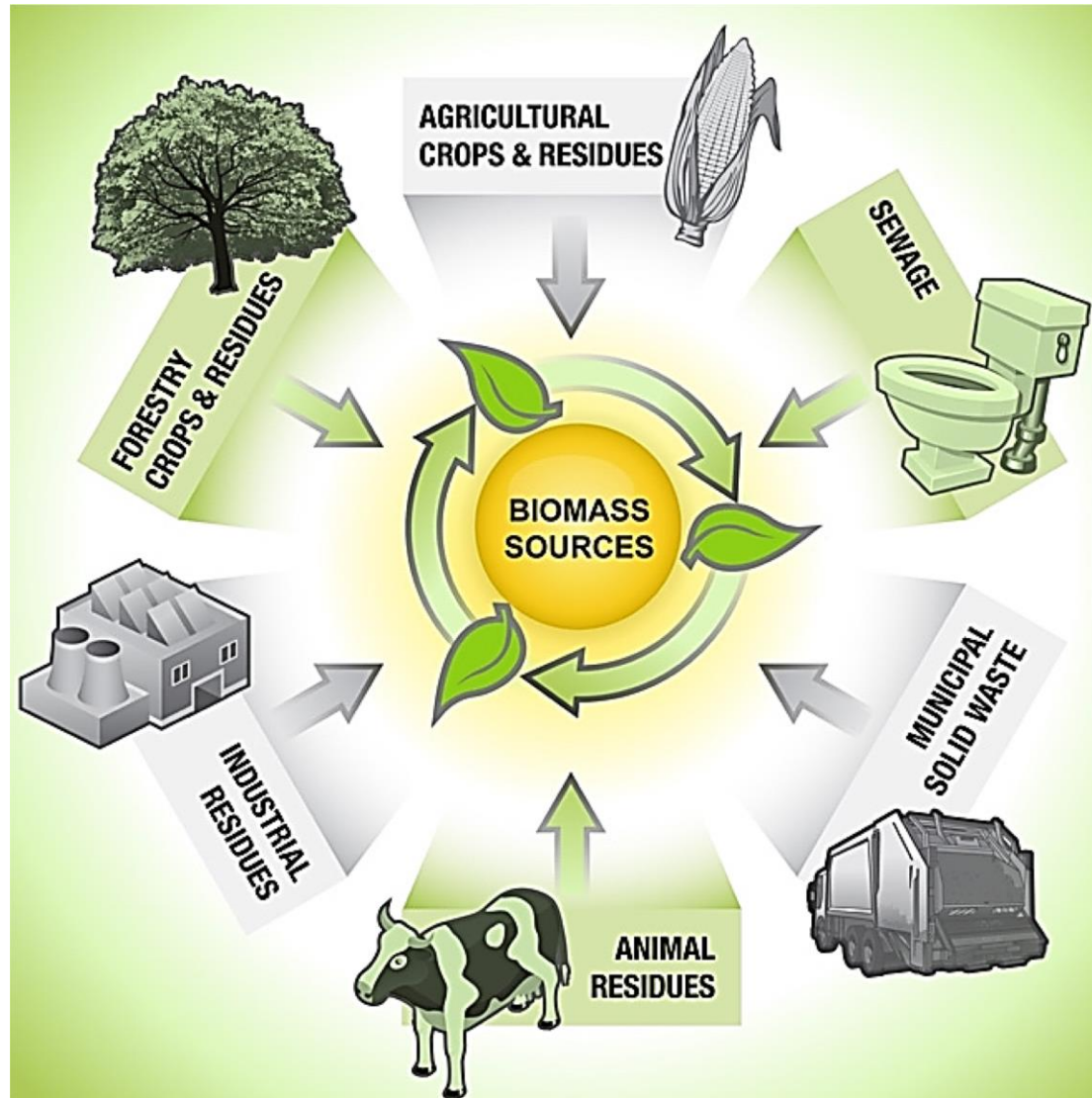
In one method, a low dam, called a barrage, is built across an inlet. The barrage has one-way gates (sluices) that allow the incoming flood tide to pass into the inlet to create a water level difference. When the tide turns, the water flows out of the inlet through huge turbines built into the barrage, producing electricity.



Energy From Biomass

- Biomass is organic matter derived from living, or recently living organisms.
- As an energy source, biomass can either be used directly via combustion to produce heat, or indirectly after converting it to various forms of biofuel. Biomass can also be converted into chemicals for making plastics and other products that typically are made from petroleum.
- Bioethanol is an alcohol made by fermentation, mostly from carbohydrates produced in sugar or starch crops such as corn, sugarcane, or sweet sorghum and widely used in USA and Brazil.

Energy From Biomass (Contd.)



Advantages of Renewable Energy

- One major advantage with the use of renewable energy is that as it is renewable therefore sustainable and will never run out.
- Renewable energy facilities generally require less maintenance than traditional generators. Their fuel being derived from natural and available resources reduces the costs of operation.
- Renewable energy produces little or no waste products such as carbon dioxide or other chemical pollutants, so has minimal impact on the environment.
- Renewable energy projects can also bring economic benefits to many regional areas.

Disadvantages of Renewable Energy

- It is not as cost-effective as other energy options because of its large capital cost and expensive storages.
- Renewable energy technologies are still significantly new to the market, meaning, they still lack the much-needed efficiency
- It may take a lot of space to install
- Renewable energy technologies totally depend on the weather (e.g., sun and wind) to be able to harness any energy. In case atmospheric conditions are not good enough, renewable energy technologies would lack the ability to generate any electricity
- Though renewable energy technologies produce no carbon emissions when they produce energy, their manufacturing, transport, and installation still creates a carbon footprint

Thank You